

SECOND PUBLIC EXAMINATION

Honour School of Physics Part B: 3 and 4 Year Courses

Honour School of Physics and Philosophy Part B

B1: FLUIDS

TRINITY TERM 2025

Thursday, 12 June, 2:30 pm – 4:30 pm

*Answer **two** questions.*

*Start the answer to each question in a **fresh book**.*

The use of approved calculators is permitted.

A list of physical constants and conversion factors accompanies this paper.

The numbers in the margin indicate the weight that the Examiners expect to assign to each part of the question.

Do NOT turn over until told that you may do so.

Page 2 contains physics formulae and data for this paper.

The questions start on page 3.

The Navier–Stokes equation for viscous, incompressible fluid flow under gravity is

$$\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{v} + \frac{1}{\rho} \nabla p + g \hat{\mathbf{z}} = \nu \nabla^2 \mathbf{v} ,$$

where $\mathbf{v}$ is the fluid velocity, ρ the density, p the pressure, g the acceleration due to gravity, $\hat{\mathbf{z}}$ the vertical unit vector and ν the kinematic viscosity.

$$1000 \text{ mb} = 1000 \text{ hPa} = 10^5 \text{ Pa}.$$

1. (a) Write down the continuity equation for a compressible fluid and derive the simpler relation that holds in the incompressible limit. Hence show that the velocity field in a two-dimensional, incompressible flow in the x - y plane can be represented by a streamfunction ψ where

$$\mathbf{v} = -\hat{\mathbf{z}} \times \nabla \psi.$$

Write down an expression for the vorticity ω in terms of ψ .

[4]

(b) State the Euler equation for two-dimensional, incompressible, inviscid flow. If the flow is unchanging with time, show that

$$\omega = \omega(\psi), \quad B = \frac{p}{\rho} + \frac{\mathbf{v} \cdot \mathbf{v}}{2} = B(\psi), \quad \text{and} \quad \underline{\omega} = -\frac{dB}{d\psi}.$$

[9]

(c) A circular vortex consists of an inner core of constant vorticity ω_0 for $r \leq r_0$ with zero vorticity elsewhere. Derive formulae for the azimuthal velocity, $v_\theta(r)$, and pressure, $p(r)$.

[6]

(d) A point vortex consists of a delta-function of vorticity. Consider two point vortices:

$$\omega = \omega_1 + \omega_2 \quad \text{where} \quad \omega_1 = A \delta(\mathbf{x} - \mathbf{x}_1), \quad \omega_2 = B \delta(\mathbf{x} - \mathbf{x}_2),$$

such that the integrated vorticity within each vortex is A and B respectively. Sketch the trajectories of the vortices in the cases $A = B$ and $A = -B$. Obtain an expression for the frequency of oscillation for the case in which the motion is periodic and where the initial separation of the vortices is L .

[6]

2. (a) Define what is meant by an *analytic function*, $w(z) = \phi(x, y) + i\psi(x, y)$. Derive the Cauchy-Riemann equations and hence show that both the real and imaginary parts of $w(z)$ satisfy Laplace's equation. Verify that the contours of ψ and ϕ are everywhere perpendicular to each other. Why are these results useful for modelling fluid flows and under what conditions? [8]

(b) Derive expressions for ϕ and ψ for the analytic function

$$w(z) = Bz^2.$$

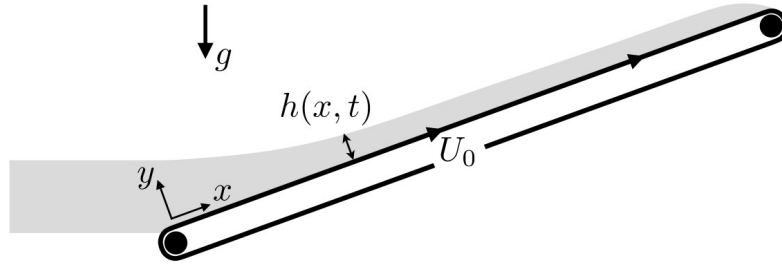
Sketch the contours of ϕ and ψ (for $-\infty < x < \infty$ and $-\infty < y < \infty$) and physically interpret the flow they represent. What is the physical significance of the point where $w'(z) = 0$? [6]

(c) Define what is meant by a *conformal mapping* $Z = f(z)$. Consider the conformal mapping

$$Z = \frac{z}{2} + \sqrt{\frac{z^2}{4} - c^2},$$

where c is real. Derive the inverse mapping, $F(z)$. Given the analytic function $w(z) = Az$, define and calculate Φ and Ψ under this mapping, and interpret the flow that they represent. Briefly discuss how and why the flow described by these functions might differ from that observed in reality. [11]

3. A thin film of viscous fluid is carried from a reservoir onto a conveyor belt moving at speed U_0 along a plane inclined at an angle θ to the horizontal. As sketched in the diagram, the thickness of the film is $h(x)$ where x is the coordinate directed along the conveyor belt and y is the coordinate perpendicular to the conveyor belt, and the film is under the influence of gravity acting in the vertical direction.



(a) Define and state the significance of the Reynolds number. Starting from the Navier Stokes equation, and stating clearly any assumptions, show that to a good approximation:

$$g \frac{\partial h}{\partial x} \cos \theta + g \sin \theta = \nu \frac{\partial^2 v_x}{\partial y^2}.$$

[8]

(b) State and interpret the appropriate boundary conditions at the conveyor belt ($y = 0$) and at the free surface ($y = h$). Hence derive a formula for the velocity profile $v_x(x, y, t)$.

[5]

(c) Suppose that the film is in an equilibrium state where its thickness is uniform sufficiently far above the reservoir, $h = h_{\text{eq}}$, and there is no net transport of fluid along the conveyor belt. Show that

$$h_{\text{eq}} = \sqrt{\frac{3\nu U_0}{g \sin \theta}}$$

and sketch the velocity profile $v_x(y)$ in this state.

[6]

(d) The conveyor belt is abruptly turned off. Sketch the velocity profile $v_x(y)$ (i) shortly after the conveyor belt is turned off and (ii) once the new boundary condition is felt throughout the film. Using dimensional analysis, estimate the time scales over which (i) the new boundary condition is felt throughout the film, and (ii) the fluid drains from the conveyor belt, assuming it initially extends a distance L along the conveyor belt.

[6]

4. (a) Consider an incompressible fluid within which the density varies linearly with height. Assuming the fluid is stably stratified, show that a fluid parcel displaced by a small vertical distance and then released will oscillate at the frequency

$$\omega = \sqrt{-\frac{g}{\rho} \frac{\partial \rho}{\partial z}}.$$

What happens if the fluid is unstably stratified?

[5]

(b) Consider two-dimensional flow in the x - z plane between two isothermal plates at $z = 0$ and $z = H$ where the temperatures are T_0 and $T_0 - \Delta T$ respectively ($\Delta T > 0$). The equations of motion are:

$$\begin{aligned} \frac{\partial v_x}{\partial t} + v_x \frac{\partial v_x}{\partial x} + v_z \frac{\partial v_x}{\partial z} + \frac{1}{\rho_0} \frac{\partial p}{\partial x} &= \nu \nabla^2 v_x, \\ \frac{\partial v_z}{\partial t} + v_x \frac{\partial v_z}{\partial x} + v_z \frac{\partial v_z}{\partial z} - b + \frac{1}{\rho_0} \frac{\partial p}{\partial z} &= \nu \nabla^2 v_z, \\ \frac{\partial T}{\partial t} + v_x \frac{\partial T}{\partial x} + v_z \frac{\partial T}{\partial z} &= \kappa \nabla^2 T, \\ \frac{\partial v_x}{\partial x} + \frac{\partial v_z}{\partial z} &= 0, \\ b &= g\alpha(T - T_0), \end{aligned}$$

where ρ_0 is a constant reference density. Write down the equilibrium solution for $T(z)$ and $p(z)$, in which there is no fluid motion. Now consider small perturbations about this equilibrium solution. Derive the vorticity equation

$$\left(\frac{\partial}{\partial t} - \nu \nabla^2 \right) \left(\frac{\partial v'_x}{\partial z} - \frac{\partial v'_z}{\partial x} \right) = -g\alpha \frac{\partial T'}{\partial x},$$

where the primes indicate small perturbations, and state the physical significance of each term in this equation. Using the remaining equations, show that

$$\left(\frac{\partial}{\partial t} - \nu \nabla^2 \right) \left(\frac{\partial}{\partial t} - \kappa \nabla^2 \right) \nabla^2 v'_z = \frac{g\alpha \Delta T}{H} \frac{\partial^2 v'_z}{\partial x^2}.$$

[11]

(c) Consider the normal mode solution, $v'_z = A \sin(kx) \sin(\pi z/H) e^{\sigma t}$. Derive an expression for σ and show that $\sigma > 0$ if

$$\frac{g\alpha \Delta T}{\nu \kappa H} > \frac{K^6}{k^2},$$

where $K^2 = k^2 + (\pi/H)^2$. Hence show that convective instability occurs if

$$\frac{g\alpha \Delta T H^3}{\nu \kappa} > \frac{27\pi^4}{4}.$$

[9]